

**Assessing the role of SUMUL in furthering Human Development in Surat and
Tapi**

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Abstract

This paper engages with the question of how cooperatives in India have been furthering the nations' development, it aims to assess the role of SUMUL in furthering socio-economic progress in the districts of Surat and Tapi. This paper investigates how has SUMUL's success and their advocacy for the marginalized communities in the tribal and rural areas of Surat and Tapi has led towards a more inclusive business model which stands on the ethos of participatory democratic model, where different stakeholders and their interests have propelled the mission of cohesive and collective development. This paper assess the impact that SUMUL has had on the health, education and livelihoods of its members, how it has engaged in dialogues and practices across different stakeholders such as the state and the banks to create solutions which benefitted everyone involved. It looks at the lessons that the cooperative model can teach us towards inculcating social progress and economic development.

Introduction

Cooperative models have been instrumental in driving the growth of India's economy, especially in rural settings because they function with the underlying agenda of inclusivity and disperse the organizational functions across various levels, such as that of the village, district and state to propel a profit making and job generating endeavour. Considering that these cooperatives are set on the premise of inclusivity and participation, one of their underlying objective for making their organizations well-equipped in terms of human capabilities, hence, human development inadvertently becomes one of their key responsibilities for smoothening their operating process, through providing longevity to the members associated (via investing in better health ecosystems), easing information dissemination (by investing in education of associated members and their successive generations, and contribute to income generation as well as availing credit facilities (as a part of their role in creating better livelihoods). The cooperative business model disperses responsibilities and obligations across various stratas of associates, this saves such entities from massive initial investment required to set-up industries, they are then able to scale up in terms of size and monetary value. Hence, the instrumental role of the cooperative model in the development of India into a self-reliant and an exceptional player in geopolitics can be

understood through the cooperatives influence in instilling and explicating the roadmap for decentralising the economic capacities, aligning the micro and meso economic development with the ‘capability approach’ propounded by Amartya Sen and Marha Nussbaum whereby the policy formation directed the citizens towards enhancing their capabilities of contribution as this model heavily relies on it, enabling inclusivity in the various production, as well as, decision making processes which has allowed the participation of members from marginalized communities through reducing the cost of investment and contributions needed for participation, as well as, embarking a shift towards entrepreneurial undertakings instead of completely relying on welfare mechanisms to fuel the development of Indian economy and society. Essentially, cooperative models are member-owned enterprises wherein individuals pool resources to meet shared economic, social and cultural requirements, they are operationalized through the premise of community welfare, voluntary participation and democratic control. Cooperatives are integral in the key industries that are responsible for India’s economic growth such as agriculture (where they provide credit, inputs and marketing support to farmers like Primary Credit Agricultural Society (PACS) and Indian Farmers Fertiliser Cooperative Limited (IFFCO)), dairy (which collect and market milk from producers like Anand Milk Union Limited (AMUL), Kerala Cooperative Milk Marketing Federation (KCMMF) and Surat Tapi District Co-operative Milk Producers’ Union Limited (SUMUL)), housing (who develop affordable housing for members), credit (that offer financial services, especially in rural areas), producer cooperatives (help artisans and small producers access markets and inputs), as well as, consumer cooperatives (which supply goods at fair prices like Kendriya Bhandar and Apna Bazar). Amongst 30 sectors, 8.4 lakh cooperative societies with 32.45 crore members across India are registered in India as of 2024, which provide 20% of all agricultural credit, 35% of all fertilizers and the cooperatives under the purview of dairy sector alone contribute 4.5% of India’s GDP.

Sector	Cooperatives	Total Members	Urban Cooperative	Urban Members	Rural Cooperative	Rural Members
Agriculture & Allied Cooperative	27,899	8,245,757	3,224	3,974,746	24,675	4,271,011
Agro Processing / Industrial Cooperative	23,580	2,227,233	11,819	1,185,090	11,761	1,042,143
Bee Farming Cooperative	349	47,136	50	8,629	299	38,507
Consumer Cooperative	23,236	10,140,478	15,298	5,793,619	7,938	4,346,859
Credit and Thrift Society	83,522	53,053,920	46,113	36,817,047	37,409	16,236,873
Dairy Cooperative	1,50,077	13,023,508	3,848	787,405	146,229	12,236,103
Educational and Training Cooperatives	1,593	4,567,440	474	1,249,948	1,119	3,317,492
Farmers	560	1,888,430	92	323,412	468	1,565,018

Service Societies (FSS)						
Fishery Cooperative	26,881	3,736,948	3,607	864,693	23,274	2,872,255
Handicraft Cooperative	5,182	91,062	2,064	37,996	3,118	53,066
Handloom Textile and Weavers Cooperative	19,557	1,635,416	8,719	722,709	10,838	912,707
Housing Cooperative Society	1,93,910	26,487,291	177,537	24,946,550	16,373	1,540,741
Jute and Coir Cooperative	71	4,746	38	3,376	33	1,370
Khadi Gramodyog	20	5,164	13	4,370	7	794
Labour Cooperative	49,569	9,096,326	16,303	3,124,486	30,266	5,971,840
Large Area Multipurpose	5,807	7,739,310	202	417,952	5,605	7,321,358

Society (LAMPS)						
Livestock Poultry Cooperative	16,951	1,127,608	1,776	138,562	15,175	989,046
Marketing Cooperative Society	10,060	3,743,335	3,083	2,250,729	6,977	1,492,606
Miscellaneous Credit Societies	7,569	5,576,590	2,049	2,526,831	5,520	3,049,759
Miscellaneous Non Credit	34,926	11,454,887	11,340	4,246,381	23,586	7,208,506
Multipurpose Cooperative	21,441	3,072,863	8,148	1,860,435	13,293	1,212,428
Primary Agricultural Credit Society (PACS)	1,02,680	130,010,362	4,664	16,753,204	98,016	113,257,158
Sericulture Cooperative	559	78,428	96	8,884	463	69,544

Social Welfare & Cultural Cooperative	2,512	685,682	1,242	382,293	1,270	303,389
Sugar Mills Cooperative	290	3,919,420	64	733,987	226	3,185,433
Tourism Cooperative	608	40,120	170	17,238	438	22,882
Transport Cooperative	4,237	136,199	1,952	83,579	2,285	52,620
Tribal-SC/ST Cooperative	2,059	1,385,145	275	546,826	1,784	838,319
Urban Cooperative Bank (UCB)	1,463	15,281,872	1,356	14,463,871	107	818,001
Women Welfare Cooperative Society	26,298	6,092,840	4,084	7,15,624	22,214	5,377,216
	8,40,446	32,45,95,516	3,29,700	12,49,90,472	5,10,766	19,96,05,044

List of Cooperatives across different sectors, number of cooperatives and their memberships in India

Source:

The initiation of cooperative society was started with the establishment of Cooperative Credit Societies Act in 1904 to curb rural indebtedness, it was expanded through a fellow legislation in 1912 which included non-credit sectors like marketing, housing and artisan support. After independence, cooperatives played an important role and were given due consideration in the five-year plans with being entrusted and empowered especially to undertake the task of agriculture and rural development. The establishment of cooperatives was made a fundamental right in 2011 with the 97th amendment to the constitution. Under the leadership of Hon. Minister Shri Amit Shah, the Ministry of Cooperation was established in July 2021, representing a significant milestone in efforts to promote cooperation among cooperatives. The Ministry's goal is to provide a distinctive administrative, legal, and policy framework to strengthen the cooperative movement. This strengthened its legislative and administrative foothold, which was furthered by the legislation of new National Cooperative Policy in 2025 which revises the previous policy which was enacted during 2002, the new policy has rolled out its vision of establishing at least one cooperative in every village by 2047, tripling the share of cooperatives in India's GDP and enrolling more than 50 crore members under their ambit, these aims are undertaken under with the intention of linking youth with entrepreneurial and employment activities. The key undertakings after the establishment of Ministry of Cooperation have been the legislation of Multi-State Cooperatives Act, 2023 which has enabled the establishment of three new national level cooperative bodies like National Cooperative Exports Limited (NECL), grain storage schemes that plans the implementation of worlds' largest grain storage through the help of cooperatives, the formation of two lack new Primary Agriculture Credit Societies (PACS) providing credit facilities and enabling the diversification of revenue generating activities by providing aid across 25 sectors including fisheries, dairy, fuel distribution, procurement etc., as well as, the foundation of the first National Cooperative University which furthers the agenda of including education regarding cooperatives in the curriculum of schools and higher education. The success of the aforementioned measures can be showcased through how 31 percent of sugar production, over 10 percent of dairy production and over 20% of

paddy procurement are undertaken by cooperatives, also more than 21 percent of business in the fishery sector are operated through cooperative societies. Such serious concerns and consideration at the national legislative level reveals the importance of cooperatives in the development of social, economic and cultural capacities of the citizens, cooperatives are defined associations amongst people that combine resources for occurring incremental benefits from an enterprise as against sole proprietorship, apart from disbursing business opportunities across different levels of members, these cooperatives also attempt to contribute in fulfilling the social objectives such as self-empowerment, mutual accord and equal participation of all.



Annual Growth Rate of Milk Production (2017-2024)

Source: (Ministry of Fisheries Animal Husbandry And Dairying, 2024)

Dairy cooperatives comprise 17.85 percent of all registered cooperatives in India with 4.01 percent of all members enrolled in cooperatives are associated with dairy cooperatives, they have played a vital role in India with regards to food security and income generation. Dairy sector has progressed at an annual rate of 6.6 percent in the last decade, India's milk processing capacity is at 126 million liters per day which is regarded as the highest in the world, dairy and animal husbandry sector contributes 4.5 per cent to the country's GDP while the dairy sector constitutes of 24 percent of the agriculture sector which is estimated to be valued at 10 lakh crore surpassing every other country in the world. Dairy cooperatives have played an explicit role in human development, especially amongst the members of rural and marginalized communities, their impact is not limited to milk production but rather plays a pivotal role in providing livelihoods, nutrition, gender equity and

social empowerment. The initiation of dairy cooperatives are often traced back to 1946, where the Gujarati village of Anand saw the beginning of the cooperative milk revolution (although, the first dairy cooperative was established in Surat back in 1939 but did not garner much success at that time). The National Dairy Development Board (NDDB) was established in 1965 with the goal of supplying the nation's growing milk demand, particularly in urban areas, and boosting the rural economy by increasing milk production. Using the Kaira District (Anand) Cooperative Milk Producers' Union Limited (AMUL) of Anand as a model, NDDB launched the Operation Flood Program in 1970 with the goal of establishing Milk Producers' Cooperatives in several likely locations throughout India. In this nation, numerous Milk Producers' Cooperatives have been established as a result of Operation Flood. The Operation Flood program has made excellent strides throughout India. A large number of Village Milk Cooperative Societies have emerged in addition to the rise in milk production and consumption per capita. There has also been a rise in the practice of keeping milk animals for scientific purposes. As a result, the amount of milk powder imported from other nations significantly decreased, and after 1976, commercial imports of the product completely ceased. But as a gift, the nation later received a small quantity of milk powder. This demonstrated that the dairy industry and the dairy development movement in India have advanced significantly since the start of the Operation Flood program (P Mahida et al., 2022).

Despite the need of the second wheat revolution in India, the feel for another white revolution is imminent and rampant across all states, this revolution is needed for instating better nutrition and health, as well as, creating more income generating endeavors (D. Mahida et al., 2018). Dairy cooperatives have played an implicit role in human development as they have provided various income generating endeavours, collective ownership and obligations have been a unique business model where performance and profits are distributed across different levels of associations, they have developed the education and health infrastructure alongside, due to the exposure and opportunities of scaling up the social ranks, they have provided the needed nutrition and aided education by incentivizing innovations across all levels of production.

	1980-81	1990-91	2000-01	2010-11	2020-21	2022-23	2023-24
Milk Procurement (in thousand kilograms per day)	2,562	9,702	16,504	26,202	55,672	59,267	66,200
Liquid Milk Marketing (in thousand kilograms per day)	2,783	8,046	13,363	21,985	36,470	42,704	43,826

Procurement and Marketing of Milk across India

Source: (NDDB, 2024)

Gujarat has been a great example of how cooperative models have fostered economic and social development, especially in the dairy sector as they helped in the realization of an intriguing avenue for income generation. The birth of dairy cooperatives in Gujarat stemmed from overcoming caste-based occupations, wherein, the formation of the cooperatives was a response to curb the role of middlemen who were providing low costs for the milk, also, the lack of quality assessment of the product was causing physical distress to the populace, hence, there were implicit implications of larger social reform at the behest of the establishment of cooperatives. As of now, Gujarat has 21 district unions, also known as federations, which have played an integral role in the dairy cooperative movement, they act as umbrella organizations for primary cooperative societies within a district and help in the facilitation of various activities such as milk procurement, processing and marketing. This paper discusses and gauges the role of one of such district union, SUMUL (Surat and Tapi District Co-operative Milk Producers Union Limited) which has been functioning across two districts in Gujarat, namely, Surat and Tapi acts as the manufacturing units for dairy products for Gujarat Co-operative Milk Marketing Federation (GCMMF), this paper attempts to trace the contribution of Sumul in the Human Development of these aforementioned districts.

Sumul: History and Role in the Development of Surat and Tapi Districts

SUMUL, originally named The Surat District Co-operative Milk Producers Union was founded on 22nd August, 1951 with the aim to counter the exploitative practices prevalent in the milk trade during that time at the hands of private traders (notably undertaken by the Bhatias), it is alleged that these traders held a monopoly over pricing, conducted malpractices of adulterating milk and denied fair returns to producers due to the lack of infrastructure and processing facilities. The name SUMUL, literally translates to “sound price” which symbolizes its undertaking towards the commitment for equitable pricing and empowerment of producers. During the inception of its establishment, it faced many challenges as the vocation of dairying was largely absent amongst the tribal communities of Surat and Tapi, there were cultural taboos that prevailed regarding buffaloes, the requisite infrastructure of roads and logistics were missing or in poor shape, as well as, the absence of knowledge about milch breeds and animal husbandry, prior to the development of dairy cooperatives the tribals kept their cows to produce bullocks rather than milk, the milk production was insufficient for even nursing the calves. In the face of the aforementioned challenges, SUMUL implemented the transformative inclusive strategies like employing Rabari women to train tribal women in milking and animal care, built cooperative buildings as community hubs, constructed roads to enable access to remote villages, as well as, introducing scientific animal husbandry and veterinary services across the rural population of Surat and Tapi districts of Gujarat, by 1983-84, over 73 percent of SUMUL’s milk came from tribal villages which was seen as a landmark achievement in inclusive cooperative development (Somjee & Somjee, 1989). SUMUL has been one of the 21 district unions under GCMMF (Gujarat Co-operative Milk Marketing Federation), which markets its products under the brand of AMUL. It had introduced fat-based payment systems in 1970 which ensured fair compensation which remunerates farmers on the account of the fat content of the milk they supply, this system incentivizes farmers to produce milk with higher fat levels because fat has been the most valuable component of milk as it is used for the production of butter, cream and cheese etc. Its success is akin to being able to develop a robust distribution network consisting of 44 milk parlors, 3,400 retail outlets and 85 contractor routes for institutional buyers.

Year	Societies	Milk Procured from societies (kg.)	Price paid per kg. fat Rs. ps	No. of Members (Societies)	No. of Trainees of Animal Husbandry Workers	No. of Artificial Insemination	No. of Pregnancy Treatment Cases	No. of Mobile Veterinary Clinics	No. of Animals Treated by Mobile Veterinary Clinics	No. of Cases of Special Veterinary Visits	Sale of Sumul Feed (M.T.)	No. of Frozen Semen Centers
1992-93	830	9,88,82,959	128.45	1,56,448	26	8,391	20,084	14	1,28,185	6,375	54,466	101
1993-94	849	11,22,80,311	132.46	1,58,088	28	11,079	22,114	16	1,51,490	8,278	56,281	65
1994-95	864	10,84,29,871	151.13	1,59,927	55	10,692	20,924	16	1,49,529	8,257	57,286	68
1995-96	900	11,69,66,093	176.75	1,95,312	22	16,815	33,239	17	1,76,397	10,258	61,668	77
1996-97	915	16,09,29,736	179.68	1,98,506	34	21,067	39,921	19	2,40,117	17,752	69,202	83
1997-98	940	18,77,96,293	184.67	2,03,906	08	22,807	49,047	19	3,13,311	21,128	82,597	89
1998-99	953	17,75,35,980	195.00	2,04,205	19	23,734	51,665	21	3,37,997	27,597	83,441	89
1999-00	960	18,56,81,310	199.00	2,05,546	23	33,505	55,216	21	3,71,334	30,596	82,264	100
2000-01	968	20,58,40,280	202.00	2,07,258	25	45,391	41,408	21	4,75,781	33,982	86,886	107
2001-02	977	19,81,62,669	205.00	2,09,208	59	52,251	73,278	21	5,14,620	39,337	96,663	115
2002-03	994	21,12,63,522	207.92	2,13,710	66	66,485	83,941	22	4,99,403	42,521	9,19,074	125
2003-04	1,006	20,73,02,286	216.00	2,17,615	70	83,765	93,326	21	5,92,200	55,220	1,01,492	140
2004-05	1,041	23,03,92,433	227.00	2,18,975	86	1,23,435	94,400	22	5,06,897	61,299	99,972	162
2005-06	1,048	25,17,46,629	233.00	2,21,057	127	1,40,120	1,18,304	22	4,48,797	79,953	1,06,491	165
2006-07	1,051	24,07,84,167	253.00	2,22,207	137	1,64,862	1,39,894	22	4,41,105	66,806	1,25,806	176
2007-08	1,055	24,33,37,021	295.00	2,23,207	106	1,94,325	1,63,495	22	4,88,628	96,602	1,15,861	201
2008-09	1,063	28,28,70,876	318.00	2,24,437	60	2,38,142	1,78,741	22	5,49,285	1,15,154	1,31,100	226
2009-10	1,068	29,50,69,162	350.00	2,25,389	80	2,70,714	2,11,487	22	6,14,56	1,36,104	1,39,254	243
2010-11	1,068	29,64,56,069	410.00	2,28,164	44	3,18,352	2,52,853	22	6,56,951	1,25,827	1,40,438	255
2011-12	1,076	32,60,83,990	485.00	2,28,924	73	3,52,381	2,91,142	22	4,74,406	1,37,585	1,50,385	262
2012-13	1,080	39,79,98,669	511.00	2,29,456	103	3,93,388	1,81,906	23	5,51,819	1,35,499	1,63,356	279
2013-14	1,095	39,03,20,107	568.00	2,30,406	121	4,01,133	3,28,547	23	4,57,741	1,26,063	1,33,709	297
2014-15	1,125	40,51,61,596	622.50	2,33,151	144	4,43,740	3,48,526	23	5,69,561	1,19,724	1,36,341	345
2015-16	1,141	50,12,38,514	691.00	2,35,301	90	4,64,305	3,61,162	26	5,44,901	1,26,639	1,34,611	398
2016-17	1,164	49,56,16,679	691.67+33.33	2,38,443	86	4,25,983	3,53,328	26	4,76,082	1,62,362	1,47,469	394
2017-18	1,173	57,53,95,510	724.17	2,40,382	67	3,08,618	2,63,137	26	4,23,373	1,52,022	1,74,256	417
2018-19	1,184	60,28,24,279	714.31	2,46,757	69	3,79,186	2,78,575	26	4,11,651	1,55,813	1,71,258	427
2019-20	1,200	53,93,04,332	756.25	2,48,469	46	3,95,244	2,46,990	26	4,74,380	1,58,108	1,97,575	434
2020-21	1,203	62,14,38,613	772.67	2,49,705	44	4,48,559	2,74,505	26	3,94,643	1,61,287	1,49,678	435
2021-22	1,208	64,88,83,629	787.00	2,50,252	44	4,55,668	2,78,283	27	3,98,381	2,69,097	1,68,794	438
2022-23	1,213	70,09,84,448.80	871.50	2,51,002	44	4,73,627	3,04,983	28	4,38,665	3,00,952	1,82,178	464
2023-24	1,215	88,46,26,340	935.56	2,51,292	22	4,85,275	3,23,832	28	4,47,124	3,34,327	1,83,319	501

Detailed Account of the Developments and Progress of SUMUL

Source: (SUMUL, 2024)

SUMUL's cooperative model has been revered for embarking multi-level development in the districts of Surat and Tapi, it has distributed around 24,679 milch animals under its Integrated Dairy Development Programme (SUMUL, 2024), while enabling its members to earn a daily income through direct bank transfers, it is estimated that its 2.5 lakh members earn 7 crore per day collectively. As of 2008, it was spread across 1068 villages (85% of which comprised tribal areas) with an average member income of 16,000 to 35,000 rupees, having its centres in Bajipura, Uchchhal, Navipardi and Nizar and procuring 29.50 liters milk annually, with an annual turnover of 4000 crores. It has engaged 4,715 women in saving groups of 71,624 women from 893 milk societies who have deposited a sum of 19,20,96,416 rupees as savings during 2021-22 providing lucrative interest rates. Under the purview of SUMUL's Milk House Aid Scheme, it has granted an aid of 80.15 lakh rupees to 35 milk societies for the construction of centres for the conduction and development of milk societies.

SUMUL has conducted various training programs pertaining to dairy entrepreneurship, governance and financial literacy amongst the members of cooperative societies, especially those hailing from marginalized communities. SUMUL has proposed an establishment of a medical hospital in the tribal areas of Surat and Tapi with a budget of 500 crore rupees investment, it has demanded state and central government for 50 percent of the funds for this endeavour. SUMUL Dairy and NDDDB intend to increase the value of dung and urine from cattle within the animal husbandry business, at the behest of joint efforts from both organizations a biogas program has led to the installation of 4743 biogas plants which has made cattle rearing families from Surat and Tapi self-sufficient in terms of their energy requirements, this scheme has been implemented with keeping various advantages in mind, such as dung being directly deposited at biogas plants prevents the spread of diseases, fermentation in the gas units destroy the seeds that produce mild and do not germinate in the soil which saves disposal cost and increases the natural fertilizing potential of dung favouring organic farming through converting organic waste into high quality manure.

Activity (2008-09)	No.
HIV control programme in rural areas	497 villages
Safe & hygienic water supply scheme	100 villages
Anemia control program for women	129 villages
Sanitation campaign	187 villages
Bio gas plant for domestic use	400 units
Tree plantation in last Five years.	32 lakh approx
Rehabilitation programme for the disabled	1397 people
Organic farming "wormy compost, demonstration and supply	330 MT
Eye camp	20504 people
Cataracts	1177
Glasses provided	12764
Plastic Surgery	145

List of Community Development Activities of SUMUL (during 2008-09)

Source: (OWFI & GKC, 2010)

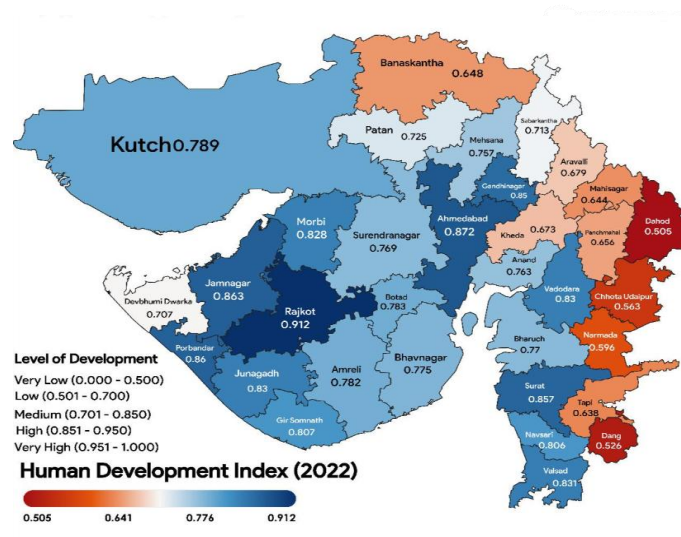
Gauging the Role of SUMUL in Human Development of Surat and Tapi Districts

SUMUL's role in creating income generating opportunities, along with its community driven initiatives towards education and healthcare have largely benefited the people hailing from Surat and Tapi districts of Gujarat, especially those from marginalized communities. All of 2,51,292 members are from the these districts

comprise of a sizable part of the population within these districts, considering that the combined population of Surat and Tapi are estimated to be around 68,88,344 (60,81,322 in Surat and 8,07,022 in Tapi as per the 2011 census), making the membership of SUMUL cooperative 3.64% of the combined population of these districts. SUMUL as a dairy cooperative keeps branching out in allied industries which are adjacent to the dairy industries, the investment made towards human capital becomes imperative in the process of creating talents which would further the initiatives undertaken by the cooperative, it has recently branched into bakery and ice-cream manufacturing to provide a wide variety of opportunities for the people of these aforementioned districts. Although, there is a lack of explicit markers of being able to determine the extent of the role that SUMUL has played in the human development of these districts, this paper attempts to assess the parameters of district level Human development report published by Gujarat Social Infrastructure Development Society under the aegis of the Government of Gujarat during 2015 and the community driven initiatives undertaken by SUMUL for the purpose to analyzing the influence that the district union has had on the development of education, health and livelihood opportunities in the region. Administratively, Surat comprises of 10 talukas (named Surat city, Mandvi, Bardoli, Palsana, Mahuva, Kamrej, Mangrol, Choryasi, Olpad and Umarpada), 5 prant, 567 gram panchayats spanning over the area of 4841 sq km while Tapi comprises of 7 talukas (named Vyara, Songadh, Valod, Uchhal, Nizar, Dolvan and Kukarmunda), 2 prant and 284 gram panchayats spanning across 3,434 sq km. The population density of Surat district is much higher than that of Tapi, as the former comprises approximately 14,000 people per square kilometre whereas the latter consists of only 234 inhabitants per square kilometre.

The presence of SUMUL is very apparent in all the 17 talukas of Surat and Tapi, this section attempts to trace the community development programs of SUMUL as well as the assesses the Human Development Indicators of both Surat in an attempt to draw understand and explicate the relationship between both, considering barring the urbanized areas of these two districts, the availability of income generating activities are scarce, especially under the consideration of the peculiar climatic conditions (semi-arid to sub-dry humid climate) and soil profile of of the region. A large part of the rural population in these districts are dependent on

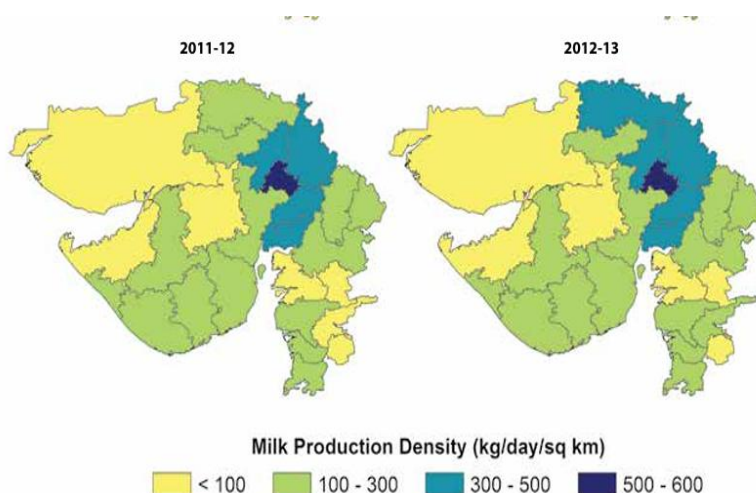
agriculture for their livelihood, dairy activities provide a suitable way for diversifying their sources of revenue. Despite the widespread presence of SUMUL dairy in Surat and Tapi districts, these districts are not amongst the top 8 districts with the highest share in milk production across the state, though both the districts are amongst the state's crossbreed cow milk production whereby Surat's contribution amounts to 198.16 thousand tonnes (comprising of 3.38%) and Tapi's contribution stands at 175.93 thousand tonnes (comprising of 3%) in the year 2022-2023 (Gurjar & Modi, 2024).



District-wise Human Development Index rating of Gujarat

Source: (Chaurasia, 2023)

Despite the rampant growth of SUMUL as a dairy cooperative, Surat and Tapi are not amongst the highest milk producing districts in Gujarat, this showcases the ample opportunity of growth amongst the rural and marginalized communities of Surat and Tapi (D. P. Mahida et al., 2018), also, these districts are not amongst the highest ranking in terms of Human Development Index (Chaurasia, 2023). Although, the recent developments and policy formation in these districts have been favouring the growth of dairy industry for the intent of increasing their consumption and export capacity of dairy products, various policies have been aiming towards providing loans, aids and grants for the purchase of milch animals in this region which are primarily targeted towards the tribal population (especially in Tapi where 84.18% of the population belong to tribal communities, as per the 2011 Census).



Milk Production Density across various districts in Gujarat

Source: (NDDB, 2014)

Education

The literacy rate of Surat is 85.53% in 2011 (showcasing a 10.88% rise from its standing of literacy rate during 2001), which is higher than the state average which stood at around 78.3% during 2011, although, the literacy rate of Tapi district stood at 68.3% in 2011 which is a significant improvement (of 11.3%) from its standing in 2001 which was at 57%. The formation of Tapi district was made during 2007 and it was carved out of the Surat district, back in 2011 a large majority of its population belong to tribal communities, the reach of education was not at par with the state average there. Similarly, many of the talukas of Surat district reflect lower literacy rates than its average, and the divide between the literacy rates between its rural and urban areas were drastically wide, the gap was 20.54% during 2001 and 14% during 2011, whereas the rural-urban divide in Tapi was even worse standing at 25.5% in 2001 and 18.2% in 2011. This poses a perplexing position for these districts with regards to socio-economic development because the lack of literacy creates issues in information dissemination preventing the populace from gaining access to information regarding policies and plans of the government as well as that of the cooperatives. Such lack of literacy especially amongst the rural and

marginalized communities are one of the plausible reasons behind why the penetration of dairy cooperatives have been limited and these districts are not at the helm of milk production in the state.

Talukas	2001				2011			
	Total	Rural	Urban	GAP	Total	Rural	Urban	GAP
Gujarat	69.14	61.29	81.84	20.54	78.03	71.71	86.31	14.60
Surat District	74.65	62.53	82.75	20.22	85.53	76.92	87.74	10.82
Olpad	73.63	73.56	74.69	1.13	83.10	82.19	87.48	5.29
Mangrol	65.35	64.26	78.03	13.77	77.46	75.04	86.20	11.16
Umarpada	52.04	52.04	*	*	67.65	67.65	*	*
Mandvi	58.88	58.88	*	*	70.78	69.27	85.37	16.10
Kamrej	63.84	63.84	*	*	78.92	78.85	81.12	2.27
Surat City	82.98	*	82.98	*	87.89	-	87.89	*
Chorasi	81.50	78.93	82.85	3.92	85.28	83.72	86.23	2.51
Palsana	67.03	63.16	79.92	16.76	80.74	76.85	86.82	9.97
Bardoli	65.79	61.14	79.71	18.57	79.67	77.15	84.57	7.42
Mahuva	69.67	69.67	*	*	81.15	81.15	*	*

* Not Applicable
Source: Census 2001 and 2011

District / Taluka	2001				2011			
	Total	Rural	Urban	Gap	Total	Rural	Urban	Gap
Tapi	57.0	54.5	80.1	25.5	68.3	66.5	84.7	18.2
Nizar*	50.1	50.1	-	-	61.1	61.1	-	-
Uchchhal*	44.2	44.2	-	-	57.5	57.5	-	-
Songadh	55.2	50.7	77.9	27.2	66.7	63.4	82.9	19.5
Vyara	62.9	59.5	82.1	22.5	74.1	72.0	86.4	14.4
Valod	63.3	63.3	-	-	74.9	74.9	-	-

*No Urban Population

Source: Census of India – 2001 & 2011, Registrar General of India

Literacy Rates Across various Talukas of Surat (left) and Tapi (right) Districts

Sources: (GSIDS, 2015, 2016)

The gap of literacy between male and female is another pertinent issue in these districts considering that the gap stands at 9.2 percent in Surat and 14.28% during 2011, there are shortcomings of latest data with regard to the literacy, especially at the district and sub-district level which makes the task of gauging trends particularly difficult. Although the available data from the previously conducted census of 2001 and 2011 reflect that despite the district averages being relatively low, the gender literacy gap between the talukas are significantly more in both Surat and Tapi districts as shown in the graphs below. Rural female literacy is substantially challenging because the cultural norms prevent women's access to formal education, while considering higher education this citation becomes worse.

Talukwise Male and Female Literacy Gap in Surat (Left) and Tapi (Right)

Sources: (GSIDS, 2015, 2016)

With regards to the enrollment percentage across Surat and Tapi being substantially high (especially at the primary level), this trend can be akin to the Sarva Sikhsha Abhiyan Mission which ensured free and fair education, this is reflected in close to total enrollment across all social groups in both the districts. This development sparks hope towards greater literacy and equality across genders and social classes in the future.

Taluka	Percentage Enrolment (6-11 age group)											
	All communities			SC			ST			Minority		
	Males	Females	Total	Males	Females	Total	Males	Females	Total	Males	Females	Total
Surat District	98.39	99.18	98.89	99.42	99.53	99.48	99.66	99.71	99.69	99.43	98.60	99.07
Olpad	99.00	99.45	99.23	100	100	100	100	100	100	100	100	100
Mangrol	99.46	99.65	99.56	100	100	100	99.86	99.87	99.87	99.83	99.98	99.91
Umarpada	95.99	98.66	97.33	97.21	98.80	98.01	99.00	99.95	99.48	100	100	100
Mandvi	98.78	99.00	98.89	100	100	100	99.95	98.95	99.45	99.58	98.63	99.11
Kamrej	100	100	100	100	100	100	100	100	100	100	100	100
Choryasi	99.95	99.88	99.92	98.99	98.65	98.82	99.56	99.99	99.78	99.65	98.66	99.65
Palsana	98.90	98.11	98.51	98.60	98.35	98.48	98.55	98.65	98.60	95.82	90.16	92.99
Bardoli	96.95	98.92	97.94	100	100	100	100	100	100	100	100	100
Mahuva	98.78	98.95	98.67	100	100	100	100	100	100	100	100	100

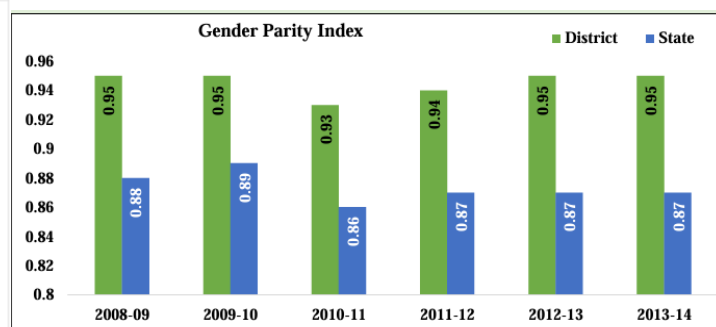
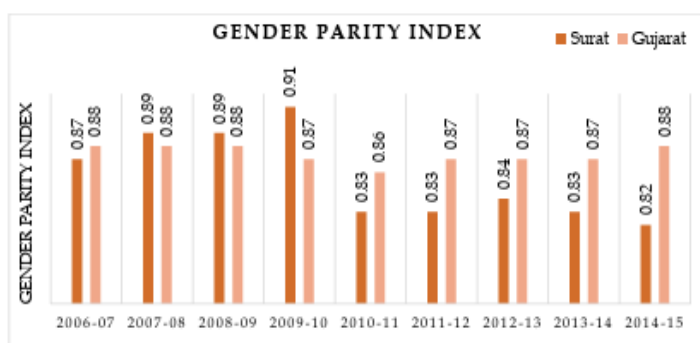
Taluka/ District	All communities			ST			Minority		
	Males	Females	Total	Males	Females	Total	Males	Females	Total
Tapi	99.88	99.91	99.89	99.88	99.89	99.88	100	100	100
Nizar	99.61	99.66	99.63	99.61	99.58	99.59	100	100	100
Songadh	100	100	100	100	100	100	100	100	100
Uchhal	99.89	99.94	99.91	99.88	99.93	99.90	100	100	100
Valod	100	100	100	100	100	100	100	100	100
Vyara	99.96	99.97	99.97	99.96	99.97	99.96	100	100	100

Source: District Primary Education Office, Tapi (Vyara)

Percentage Enrollment (between 6-11 age groups) across Surat (Left) and Tapi (Right)

Sources: (GSIDS, 2015, 2016)

The gender parity in both the districts reflect a positive trend towards equality in Surat but the disparity in Tapi is has been constantly high without any signs of narrowing, this reflects a deeper issue while considering the social demographics of Tapi which is mostly populated by marginalized and tribal communities, the efforts of expanding the reach and access to education amongst women haven't been very successful. This is one of the gaps that SUMUL's community initiatives have tried to address because inculcating women under the purview of dairy cooperatives holds the potential of multiple problems that rise of membership and milk production face in these district, considering that migration becomes a pivotal issue and prevents rural development due to the brain drain towards urban areas in search of more remunerative activities often leaving women and children behind, involving and inculcating women as entrepreneurs and members in dairy cooperatives hold the possibility of significantly rising the milk production and income generating activities in these districts.



Gender Parity Index across Surat and Tapi Districts

Sources: (GSIDS, 2015, 2016)

SUMUL has conducted 195 dedicated programs reaching to 12,979 women across the villages of Surat and Tapi districts, these programmes largely focus on informing and educating women about financial literacy, cooperative governance, entrepreneurship and self-employment. These programs have been designed and conducted with the intent of empowering women to participate in decision making, manage saving groups and leading dairy education. Through the educational programs for women, SUMUL envisions imparting the ability of transforming households by boosting income, confidence and community leadership. SUMUL also delivers structured training to committee members from village-level milk societies frequently, with the emphasis towards informing the trainees about the best governance practices, record keeping and transparency, as well as, instilling a practice of member engagement and grievance redressal, such training are conducted for building institutional capacity and democratic functioning at the grassroot levels of the cooperatives. Under the purview of Progressive Entrepreneurship Development Program, SUMUL organizes a 5-day training camp for high-yielding milk producers across the societies under its ambit, this programme equips the selected trainees with technical manuals, provides exposure visits to dairy farms that are successful and facilitate monthly on-site consultancy by experts, this program has benefitted 331 trainees during 2022-23 and 382 trainees during 2023-24. SUMUL also conducts educational field visits to its manufacturing plant for school and college studies, during these visits students are taught about modern dairy processing, packaging and hygiene standard, as well as sustainable practices and automation with the mission of bridging the gap between classroom learning and real-world applications which would inspire more innovative acumen amongst students. The Integrated Dairy Development Programme (IDDP) involves training subsidies for members belonging to tribal communities or those below poverty lines, these trainings educate members about animal care, feed management and financial planning. Though the evidence of SUMUL's involvement in providing formal education or scholarships and internships to students have not been documented per se, its approach holds an implicit commitment towards holistic empowerment of the members associated with it.

Health

The public health infrastructure follows a three tier structure, following the mandate of National Health Policy, 1983, whereby the first tier comprises of an Auxiliary Nurse Midwife (ANM) and a male multipurpose worker (MPW) for the population of 5,000, the intermediary tier consists of Sub Centre (SC) which provides primary health care interventions (substantially focussing of maternal and child health) and on the top of which are Primary Health Centres which are referral unit for 6 subcentres (having 4-6 beds and headed by a medical officer in-charge) and 14 subordinate paramedical staff, at the top of this tiered system are Community Health Centres which have 30 bedded referral unit/hospital for 4 public health centres that are under its supervision. The health infrastructure in the districts of Surat consists of 14 community health centres, 337 sub-centres. The aforementioned tables provide a detailed account of the public health infrastructure in Surat and Tapi.

Taluka / District	Infrastructure for Healthcare							
	CHC	PHC	Sub-Centres	Private-Hospitals	Dispensaries	Beds	Doctors	Nurses
Surat District	14	50	337	112	50	720	106	448
Olpad	2	7	37	5	7	102	15	63
Mangrol	2	6	45	9	6	96	14	56
Umarpada	1	3	18	0	3	48	7	28
Mandvi	2	7	60	7	7	102	15	63
Kamrej	2	5	32	9	5	90	13	49
Choryasi	1	5	19	42	5	60	9	42
Palsana	1	4	25	10	4	54	8	35
Bardoli	1	7	54	28	7	72	11	56
Mahuva	2	6	47	2	6	96	14	56

District / Talukas	Numbers			Per 3000 Population	Per 20000 Population	Per 80000 Population
	SCs	PHCs	CHCs	SCs	PHCs	CHCs
Tapi	235	30	5	0.87	0.74	0.50
Nizar	41	5	1	0.95	0.77	0.62
Uchchhal	28	4	1	0.95	0.90	0.90
Songadh	62	8	1	0.81	0.70	0.35
Vyara	74	9	1	0.83	0.67	0.30
Valod	30	4	1	0.99	0.88	0.88

Public Health Infrastructure of Surat (left) and Tapi (right) district (as of December 2014)

Source: (GSIDS, 2015, 2016)

Despite the robust health infrastructure in both the districts, there are various challenges pertaining to access to quality nutrition within these districts as evidenced by the statistics mentioned below, as only 3.2% of the children aged between 6-23 months receive an adequate diet in Surat and 6.7% in Tapi. The lack of nutrition results in stuntedness amongst children 36.1% in Surat and 41.7% in Tapi, as well as 32.5% children under the age of 5 in Surat and 51.8% in Tapi are underweight. This reflects an acute crisis of the lack of proper diet and nutrition amongst people from these districts, milk production does not only serve the purpose of income

generation but also hold the potential to provide better nutrition and diet amongst the people of the districts considering the high nutritional value and milk products have, especially for children and adolescents.

Indicators	NFHS-5 (2019-20)	NFHS-4 (2015-16)
Child Feeding Practices and Nutritional Status of Children	Total	Total
67. Children under age 3 years breastfed within one hour of birth ¹⁵ (%)	44.1	46.0
68. Children under age 6 months exclusively breastfed ¹⁶ (%)	*	*
69. Children age 6-8 months receiving solid or semi-solid food and breastmilk ¹⁶ (%)	*	*
70. Breastfeeding children age 6-23 months receiving an adequate diet ^{16, 17} (%)	1.9	0.0
71. Non-breastfeeding children age 6-23 months receiving an adequate diet ^{16, 17} (%)	*	*
72. Total children age 6-23 months receiving an adequate diet ^{16, 17} (%)	3.2	0.0
73. Children under 5 years who are stunted (height-for-age) ¹⁸ (%)	36.1	30.0
74. Children under 5 years who are wasted (weight-for-height) ¹⁸ (%)	26.0	26.2
75. Children under 5 years who are severely wasted (weight-for-height) ¹⁹ (%)	14.7	8.1
76. Children under 5 years who are underweight (weight-for-age) ¹⁸ (%)	32.5	36.1
77. Children under 5 years who are overweight (weight-for-height) ²⁰ (%)	5.3	2.9
Nutritional Status of Women (age 15-49 years)		
78. Women whose Body Mass Index (BMI) is below normal (BMI <18.5 kg/m ²) ²¹ (%)	21.0	18.4
79. Women who are overweight or obese (BMI ≥25.0 kg/m ²) ²¹ (%)	25.1	34.5
80. Women who have high risk waist-to-hip ratio (≥0.85) (%)	58.3	na
Anaemia among Children and Women		
81. Children age 6-59 months who are anaemic (<11.0 g/dl) ²² (%)	83.6	42.3
82. Non-pregnant women age 15-49 years who are anaemic (<12.0 g/dl) ²² (%)	69.3	39.4
83. Pregnant women age 15-49 years who are anaemic (<11.0 g/dl) ²² (%)	(58.7)	(29.0)
84. All women age 15-49 years who are anaemic ²² (%)	69.0	39.0
85. All women age 15-19 years who are anaemic ²² (%)	76.2	39.9

Nutrition and Health Statistics of Surat

Source: (MHFW, 2021)

Indicators	NFHS-5 (2019-20)	NFHS-4 (2015-16)
Child Feeding Practices and Nutritional Status of Children	Total	Total
67. Children under age 3 years breastfed within one hour of birth ¹⁵ (%)	46.8	35.9
68. Children under age 6 months exclusively breastfed ¹⁶ (%)	*	*
69. Children age 6-8 months receiving solid or semi-solid food and breastmilk ¹⁶ (%)	*	*
70. Breastfeeding children age 6-23 months receiving an adequate diet ^{16, 17} (%)	(4.0)	7.4
71. Non-breastfeeding children age 6-23 months receiving an adequate diet ^{16, 17} (%)	*	*
72. Total children age 6-23 months receiving an adequate diet ^{16, 17} (%)	6.7	9.4
73. Children under 5 years who are stunted (height-for-age) ¹⁸ (%)	41.7	35.9
74. Children under 5 years who are wasted (weight-for-height) ¹⁸ (%)	36.6	35.8
75. Children under 5 years who are severely wasted (weight-for-height) ¹⁹ (%)	17.1	9.6
76. Children under 5 years who are underweight (weight-for-age) ¹⁸ (%)	51.8	42.4
77. Children under 5 years who are overweight (weight-for-height) ²⁰ (%)	1.9	1.2
Nutritional Status of Women (age 15-49 years)		
78. Women whose Body Mass Index (BMI) is below normal (BMI <18.5 kg/m ²) ²¹ (%)	35.4	43.4
79. Women who are overweight or obese (BMI ≥25.0 kg/m ²) ²¹ (%)	12.6	8.8
80. Women who have high risk waist-to-hip ratio (≥0.85) (%)	57.7	na
Anaemia among Children and Women		
81. Children age 6-59 months who are anaemic (<11.0 g/dl) ²² (%)	80.7	49.5
82. Non-pregnant women age 15-49 years who are anaemic (<12.0 g/dl) ²² (%)	77.6	54.3
83. Pregnant women age 15-49 years who are anaemic (<11.0 g/dl) ²² (%)	(72.0)	*
84. All women age 15-49 years who are anaemic ²² (%)	77.4	54.4
85. All women age 15-19 years who are anaemic ²² (%)	72.8	57.8

Nutrition and Health Statistics of Tapi

Source: (MHFW, 2021)

SUMUL has undertaken various programmes for combating malnutrition in the region, few of the programmes that it has implemented for redistribution of goods are (i) Food Factory Take Home Ration, which

was a facility established by the collaborative efforts of SUMUL, Amul and Integrated Child Development Scheme, whereby the union sold THR products worth 301.68 crores, and (ii) as a part of Gujarat Government's Tribal Sub Plan and Department of Integrated Child Development Scheme, SUMUL has been participating in "Dudh Sanjeevani Yojana" which provide milk to children attending ashramshalas, primary schools and anganwadis (i.e., day care centres) in Surat and Tapi districts. SUMUL has expanded the health coverage of its members under Pradhan Mantri Bima Yojna (Jain, 2025).

No. of. 200ml pouch given in Surat-Tapi District during the year	Total milk given during the year	Granted amount by Government during the year (Rs.)
3,09,70,450	61,94,090.00	32,34,04,554.54

Milk Distributed amongst Anganwadis, Ashramshalas and Primary Schools

Source: (SUMUL, 2024)

Income

While the working population increased from 61% in 2011 to 64% in 2021, youth participation in economic activities decreased to 37% in 2022, according to the India Employment Report (2024). Furthermore, according to the Centre for Monitoring Indian Economy (CMIE), the unemployment rate increased to 9.2% in June 2024, despite the labor participation rate increasing to 41.4%. Cooperatives emerge as a workable option in this situation, providing long-term job opportunities and bolstering public and private sector initiatives. Cooperatives are crucial for promoting employment, entrepreneurship, and skill development, all of which are necessary for economic recovery through putting an emphasis on shared responsibility. The idea of inclusive business has been lucrative for a variety of reasons, largely because of the opportunities to scale from a production and retail perspective.

Taluka / District	Cultivators		Agricultural Labourers		Household Industries		Other workers		Total Main Workers
	Number	Percentage of Total Main Workers	Number	Percentage of Total Main Workers	Number	Percentage of Total Main Workers	Number	Percentage of Total Main Workers	
Surat	101747	4.23	274736	11.42	32933	1.37	1995872	82.98	2405288
Olpad	13169	16.39	26631	33.14	618	0.77	39941	49.70	80359
Mangrol	11271	14.75	29307	38.34	645	0.84	35210	46.07	76433
Umarpada	7254	22.42	21769	67.28	346	1.07	2986	9.23	32355
Mandvi	25074	30.05	39631	47.50	1398	1.68	17330	20.77	83433
Kamrej	7888	10.48	32631	43.34	735	0.98	34034	45.21	75288
Surat City	7135	0.41	13261	0.77	26159	1.51	1683588	97.31	1730143
Choryasi	4744	4.35	10900	9.99	742	0.68	92672	84.97	109058
Palsana	4248	6.67	20964	32.90	789	1.24	37715	59.19	63716
Bardoli	7816	8.38	44667	47.87	960	1.03	39869	42.73	93312
Mahuva	13148	21.49	34975	57.16	541	0.88	12527	20.47	61191

Source: Census 2011

Occupation Wise Proportion of Main Workers in Surat

Source: (GSIDS, 2015, 2016)

District / Taluka	P/M/F	Cultivators		Agri. Lab.		HH Inds.		Others	
		2001	2011	2001	2011	2001	2011	2001	2011
Tapi District	Persons	44.35	23.45	50.01	56.51	2.79	2.15	27.97	17.90
	Male	42.04	29.02	34.19	47.03	2.25	1.83	29.73	22.13
	Female	48.31	16.45	77.10	68.41	3.71	2.55	24.97	12.59
Nizar	Persons	29.71	13.58	79.14	79.48	0.90	0.36	19.06	6.59
	Male	34.38	19.24	57.49	71.65	0.95	0.42	18.01	8.69
	Female	21.95	5.89	115.03	90.11	0.78	0.27	20.80	3.73
Uchchhal	Persons	37.41	14.61	69.99	76.99	1.50	1.18	16.35	7.22
	Male	34.84	18.85	52.48	70.67	1.26	1.23	18.84	9.26
	Female	41.07	10.04	94.92	83.80	1.84	1.12	12.82	5.03
Songadh	Persons	52.67	22.93	38.99	52.23	3.58	2.48	33.55	22.36
	Male	45.97	28.53	24.27	41.09	2.77	2.19	36.35	28.20
	Female	65.15	16.08	66.39	65.84	5.11	2.84	28.35	15.23
Vyara	Persons	54.25	33.47	38.12	42.21	3.73	2.64	30.62	21.69
	Male	49.44	39.79	23.46	31.54	2.91	2.37	32.01	26.30
	Female	63.18	25.56	65.29	55.55	5.25	2.98	28.04	15.92
Valod	Persons	23.98	17.36	54.40	57.13	1.90	3.36	29.36	22.15
	Male	27.36	22.06	44.48	50.08	1.56	1.88	31.03	25.98
	Female	19.10	10.50	68.73	67.44	2.39	5.52	26.94	16.54

Source: Census-2011

Occupation Wise Proportion of Main Workers in Tapi

Source: (GSIDS, 2015, 2016)

The aforementioned tables suggest that most of the workers are engaged in agricultural activities, either as cultivators who have their own land or agricultural labourers, especially in the predominantly rural talukas of the Surat and Tapi. Dairy is considered as one of the primary industry pertaining to agriculture and animal husbandry, although, there is no clear demarcation of how many people are directly involved in the dairy industries within Tapi and Surat, although, the assumption of the number of SUMUL cooperative members being directly or indirectly involved in the dairy industry is often taken in stride, despite the lack of clarity regarding how many of the 2.5 lakh members are actively engaged with this industry. There is substantial milk production in this region as the following table suggests which refers to the data shared by the National Bank

for Agriculture and Rural Development, this study was conducted to gauge the feasibility and potential of the dairy industry as a plausible revenue generating source within the region. This study highlights some key assessments about the dairy industry.

i)	Per day milk production	1250 tonnes
ii)	Number of Milk Collection Centre	1547
iii)	Number of Bulk chilling Units	549
iv)	Number of bulk milk cooling units	03
v)	Number of village dairy cooperative societies	489
vi)	Number of milkoscan(FT 120)	8
vii)	Number of milk producers covered under cooperative fold	238443
viii)	Number of private dairies operating and their coverage	NA
ix)	Per day procurement by Cooperative dairies / Private dairies	NA
x)	Adequacy of market procurement infrastructure	There are 675 milk societies and 03 bulk milk cooling units in the districts.

Details of SUMUL's features in Surat and Tapi¹

Source: (NABARD, 2024)

SUMUL dairy has 1547 Automatic Milk Collection Centre, 549 Bulk Chilling units, 1907 Automatic Milk Collection Units with computerized counters for milk procurement in Surat and Tapi District. The information shared herein explicates the feasibility of dairy as plausibly profitable endeavour at the behest of the infrastructure that SUMUL has established in the region, the aforementioned table suggests that this region holds to requisite infrastructure for dairy farmers availing them to conveniently sell their produce.

Income		Year 1	Year 2	Year 3	Year 4
Price per Lt	29				
Lactation days (2 animals)		430	520	550	540
Avg. Milk Yield per day(Lt)	12				
Income from sale of milk		1,49,640	1,80,960	1,91,400	1,87,920
Dry Days		110	200	170	180
Expenses					
Cost of feeding					
Lactation days		77,400	93,600	99,000	97,200
Dry Days		10,450	19,000	16,150	17,100
Veterinary expenses		2,500	2,500	2,500	2,500
Insurance		6,000	6,000	6,000	6,000
Miscellaneous		2,000	2,000	2,000	2,000
Total Expenses		98,350	1,23,100	1,25,650	1,24,800
Gross Surplus		51,290	57,860	65,750	63,120

Estimated Yearly Income from small scale dairy consisting of two milch animals

Source: (NABARD, 2024)

¹ The number of members in this table are 2,38,443 because NABRAD study was conducted till 2023, the newly added members are not taken into account here.

An intriguing development of the study conducted by NABRAD suggests that returns from small scale dairy farms are incremental yet low, considering how labour and capital intensive maintaining milch animals can be, despite being profitable, the returns for small scale farmers having two milching animals might act as way of disincentive for people to adopt dairy but it still showcases the potential it may hold for rural household for diversifying their household incomes through engaging in this industry. This study suggests that procuring loans individually might be difficult but as a cooperative and societies dairy farmers can collectively demand loans that provide a more profitable outlook for the banks, in terms of better margins through interests. The formation of cooperatives, hence, adds another advantage because most small scale farmers might lack the collateral or financial means of garnering loans, considering that many of the citizens in this region belong to marginalized communities.

		Year 1	Year 2	Year 3	Year 4
1. Capital Costs		1,31,000	0	0	0
2. Recurring costs		82,700	1,02,500	1,04,900	1,04,100
Total Costs		2,13,700	1,02,500	1,04,900	1,04,100
Total Benefit		1,49,640	1,80,960	1,91,400	1,87,920
Surplus		-64060	78460	86500	83820
Discounting factor at 15%		0.870	0.756	0.658	0.572
NPV of costs @ 15%		1,85,826	77,505	68,973	59,520
NPV of benefits @ 15%		1,30,122	1,36,832	1,25,849	1,07,444
BCR		1.28			
IRR		114%			

Cost Benefit Analysis of Small Scale Dairy Operations

Source: (NABARD, 2024)

While conducting the year on year growth of small scale dairy industries, the NABRAD study suggests that the profitability of very small scale dairy farms can only be attained in the second year of its operations, also, the first year is the most expensive because it entails the cost of procuring milching animals which is substantial portion of the operational costs. The sales of milk produce cannot make up for the costs incurred during the initial year of establishing a small-scale dairy (consisting of only 2 milching animals). Hence, aligning with cooperatives provide dairy farmers the leverage and power of collective bargaining for negotiating

the terms of their returns of the debts they may incur in establishing the dairy in the first place. Despite the benefit cost ratio and internal rate of return being lucrative, they are not met in the first year of establishment. The means of availing social mobility through the dairy industry can only be met in relatively larger dairies, which further increase the initial investments. Due to the democratic model of cooperatives and the commitment of SUMUL towards grievance redressal, it has introduced Integrated Dairy Development Programme (IDDP) which has distributed 24,679 milch animals amongst member belong to the tribal communities and families that are below the poverty line, this programme has enabled the entry for many cooperative members into dairy farming by providing 72.63 crore in subsidies while also providing an assistance of 45,000 (covering the animal insurance premiums for three years, SUMUL dan which offers feed to the farmers and training expenses)..

Details	2022-2023	2023-2024
Purchasing of Animals	1,101	2,031
Subsidy Amount (in Lakhs)	495.19	913.95
Loan Amount (in Lakhs)	247.67	456.35

Grants, Loans and Aid provided by SUMUL for incentivizing entry into dairy farming

Whereas back in 2018-19, the government of Gujarat offered various schemes for promoting the establishment of dairy farms through subsidies, this scheme required the prospective beneficiaries to meet three requirements, namely, (i) the construction of cattle shed (at least 600 square feet), purchase of 12 milching animals and the payment of insurance premium. For fulfilling the aforementioned requirements, 1.5 lakh rupees was provided in subsidy (which would be half of the total investment), 8.5% loan interest subsidy for five years of loan taken for this project as well as, a subsidy of Rs. 43,200 was availed to the beneficiaries for the payment of insurance premiums. Also, this scheme provided more subsidies on the paraphernalia required for conducting dairy farming, like 18,000 rupees on the purchase of a chaff cutter, 33,750 rupees for milking machine and

9,000 rupees for a fogger system. A sum of 3,06,350 rupees were designated for scheduled caste and women beneficiaries, this scheme was successful in incentivizing the installation of 16 dairy units across Surat and Tapi districts. As a part of NABARD's Dairy Entrepreneurship Development Scheme (DEDS), it offers low-interest loans and capital subsidies to prospective dairy farmers; this scheme has financed more than 2,000 animal units with the help of 2.41 crore in subsidies.

Conclusion

Cooperatives are garnering attention in legislation and markets, they hold great potential for creating collective ownership models which are inclusive, inquisitive and democratic in their processes. Such cooperatives have become prominent in various sectors and are instrumental in challenging monopoly. These cooperatives are beneficial for both producers and consumers on account of the price mechanisms that they adopt which provide a fair remuneration for the producers and fair price for consumers. Dairy cooperatives across India have been garnering much attention and success as they are able to avail collective bargaining with various stakeholders, providing the requisite funds, aids and debts to the participants to undertake their endeavors. Due to the collective nature of their operations, they are rooted in the commitment towards nurturing talent and investing in human capital, their implicit endeavours of providing better health facilities for the longevity and better living condition of their members, providing opportunities for learning and availing income generation for its members have strengthened their position as engines of social and economic progress in India. One such dairy cooperative SUMUL which is functional in the southern districts of Gujarat have been very successful in providing better opportunities for its members, in terms of education, health and livelihoods, it has been successful in bringing attention amongst all stakeholders towards the potential of dairy farms, they cooperative model has been an exemplar of furthering socio-economic development in a region which was marked by exclusion, they have integrated technologies in ways that better the lives of the communities that it engages with.

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